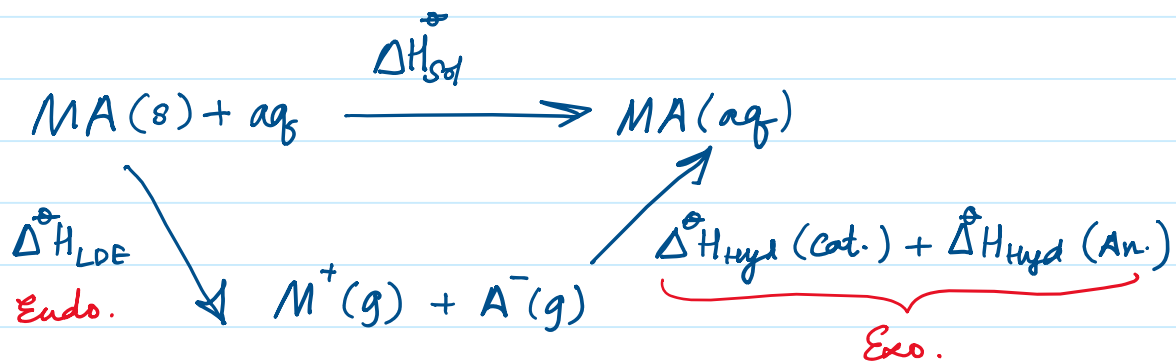


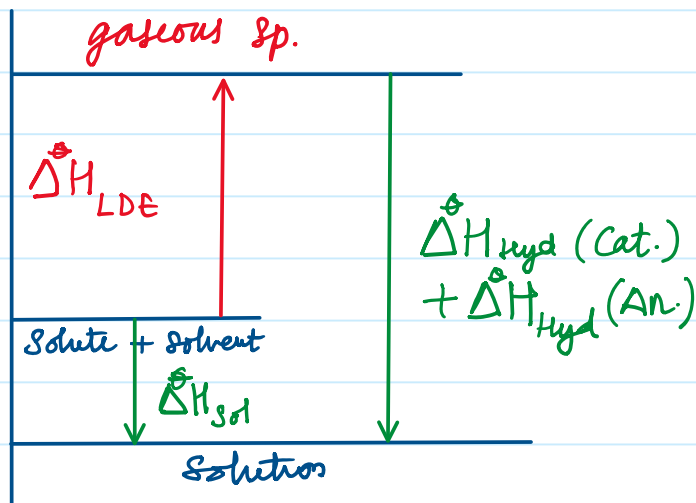


latt. dissoc. enth.

e.g. $\overset{\circ}{\Delta}H_{\text{sol}}(\text{NaCl}) = \overset{\circ}{\Delta}H_{\text{LDE}}(\text{NaCl(s)}) + \overset{\circ}{\Delta}H_{\text{hyd}}(\text{Na}^+(\text{g})) + \overset{\circ}{\Delta}H_{\text{hyd}}(\text{Cl}^-(\text{g}))$

1. Dissolution is entropically favored $\because$ solution is more disordered than liquid & solid.
2. Hydration is not entropically favored $\because$ water molecules lose freedom of movement (become more ordered) because they form interactions with ions.
3. $\overset{\circ}{\Delta}H_{\text{LDE}}$ is always positive.

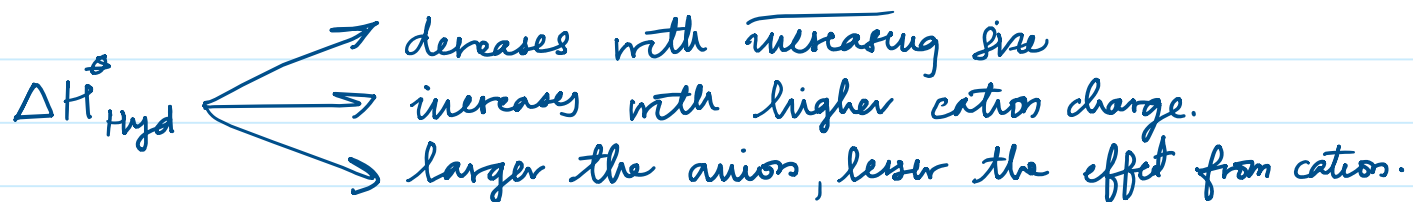
one case.



But,

If $\overset{\circ}{\Delta}H_{\text{LDE}} > \sum \overset{\circ}{\Delta}H_{\text{hyd}}$, excess energy is needed from outside. Until then the reaction would not proceed.

$\therefore$ Enthalpy of solvation is negative



Halogen redox reactions



I₂ is nonpolar and resists dissolving in H₂O (very polar).

In the presence of excess I⁻(aq),

